Unit - 4 WSN

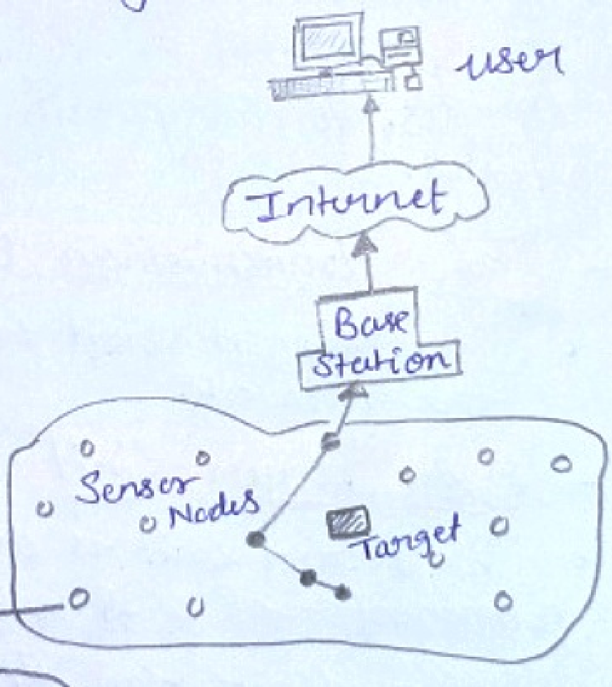
important topics

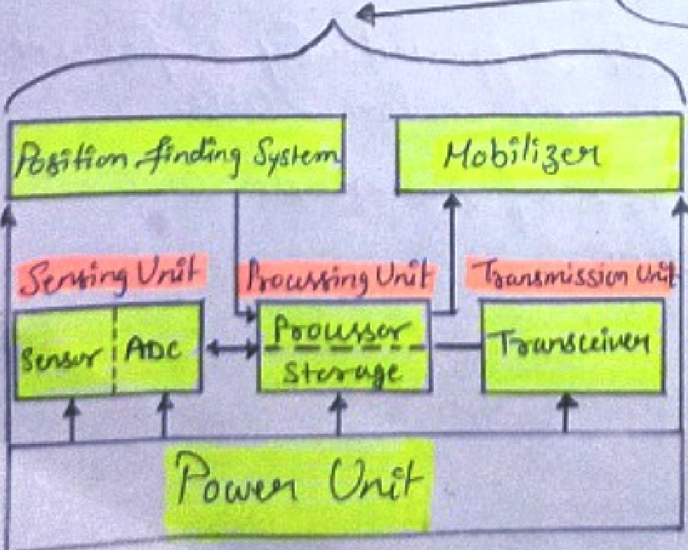
- ☑ Sensor Node Hardware
- ☑ Berkeley Mote
- ☑ Programming Challenges
- ☑ Node-Level Software Platforms & Simulators.
- ☑ Security Issues
- ☑ Future Research Direction

Sensor Node - Hardware Components

⇒ A Sensor Node is made up of four basic Components namely :-

- Sensing Unit
- Processing Unit
- Transceiver Unit
- A power Unit





Components of Sensor Nodes

Power generator

Hardware - Sensing Unit

- The Sensing Units are usually composed of two subunits -
- Sensors and Analog to Digital Converters (ADCs).
- The analog signals produced by the sensors are converted to digital signals by the ADC, and then fed into the processing unit.

- The Processing Unit

- The Processing unit is generally associated with a small storage unit and
- It can manage the procedures that make the sensor node collaborate with the other nodes, to carry out the assigned sensing tasks.

- The Transmission Unit

- The Transmission unit connects the node to the network.

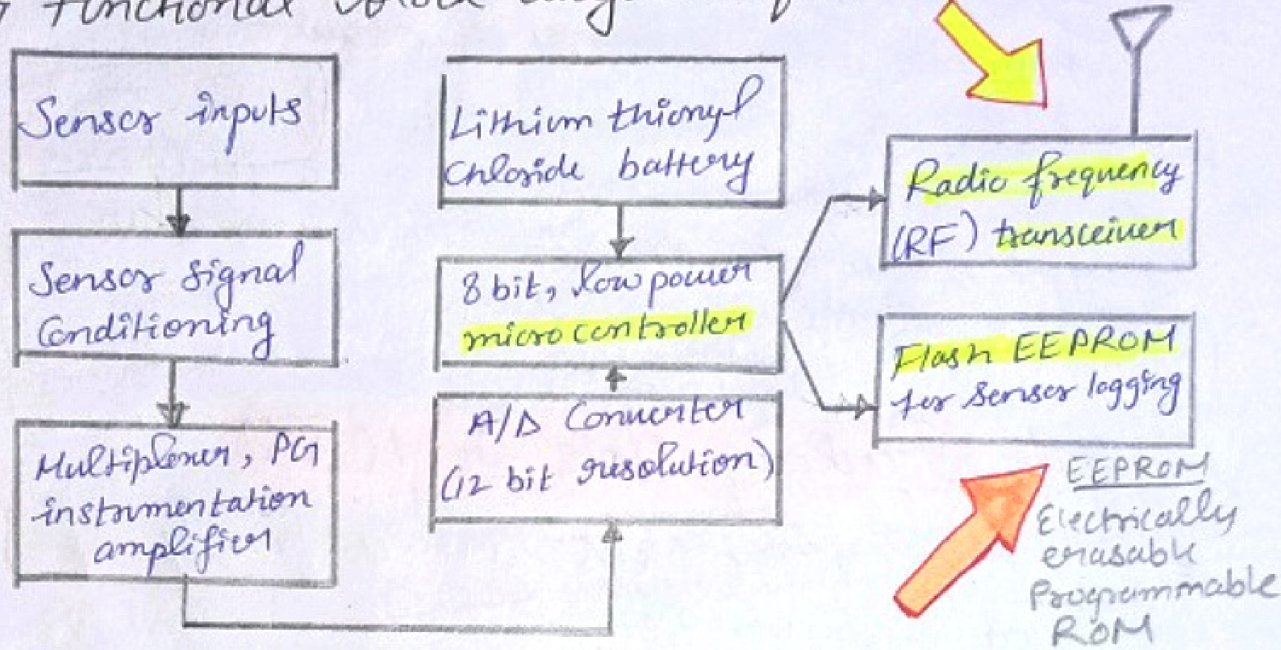
- The Power Unit

- The Power unit is one of the most important components of a sensor node.
- Power Units can be supported by a power scavenging unit such as solar cells.
- The other subunits, of the node are application dependent.

Operation of Sensor Node

- In order for a WSN to operate properly, the sensor nodes require an operating system, a routing protocol and a simulator.

⇒ Functional block diagram of Sensor Node is :-



- The radio link may be swapped out as required for a given applications' wireless range requirement and the need for bidirectional communications.
- Using flash memory, the remote nodes acquire data on command from a base station, or
- By an event sensed by one or more inputs to the node.
- Moreover, the embedded firmware can be upgraded through the wireless network in the field.
- The Microcontroller has a number of functions as listed below,
  → Managing data collection from the sensors.
  → performing power management functions.
  → Interfacing the sensor data to the physical radio layer.
  → Managing Radio N/W Protocol.

⇒ A key aspect of any wireless sensing node is to minimize the power consumed by the System.

⇒ Usually, the radio Subsystem requires the largest amount of power.

⇒ Therefore, data is sent over the radio network only when it is required.

⇒ Thus, the hardware should be designed to allow the microprocessor to judiciously control power to the radio, sensor, and sensor signal conditioner.

Berkeley Mote in WSN

⇒ The Berkeley Motes are a family of embedded sensor nodes.

"nodes developed by researchers in the university of California, Berkeley"

"Mote literally means a tiny piece of Something"

Motes

⇒ They are tiny, Self-Contained, battery powered computers with radio links,

⇒ which enable them to communicate and exchange data with one another, and to self organize into AD-HOC networks.

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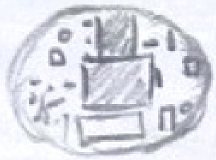
Berkeley Mote consists of :-

- Micro-Controller with internal flash program memory.
- Data SRAM (Static RAM)
- Data EEPROM.
- A set of Actuator and sensor devices, inc. LEDs
- A low power transceiver.
- An analog photo sensor.
- A digital temperature sensor.
- A serial port.
- A small coprocessor unit.



Ques. What are the key hardware components and features of Berkeley Motes that have facilitated their widespread use in Wireless Sensor Networks (WSN).

Ans→ Core Components :-

1. **Microcontrollers (MCU)** : The brain of the mote, responsible for processing data, executing the software, and controlling other components. Early Berkeley Motes, like the MICA series, often used ATmega microcontrollers, known for their low power consumption and sufficient processing capabilities for sensor data management.

MICA Series :- MICA is a platform of WSN devices developed by UC Berkeley's research group on wireless sensors

2. **Radio Transceiver** :- Enables wireless communication between sensor nodes and with a base station. The choice of radio technology impacts the mote's range, data rate, and energy consumption.

Eg→ MICA mote uses 2.4 GHz IEEE 802.15.4 compliant transceiver

3. Sensors :- Berkeley Motes can be equipped with a variety of sensors to measure environmental parameters such as temperature, humidity, light, vibration, and more. The modularity of the design often allows for easy integration of different sensors depending on the application needs.

modularity
↓
पूरा कैसा
होता है।

4. Power Source :- Typically powered by batteries, with a design focus on minimizing energy consumption to extend operational life. Some motes are designed to work with energy harvesting technologies, like solar power, to further prolong their deployment without maintenance.

5. Memory :- Includes both RAM for temporary data storage during operation and flash memory for storing the operating system, application code, and collected data. The amount of memory determines how much data can be processed and stored locally on the mote.

6. Interface Ports :- For programming the device, debugging, or connecting additional peripherals. Early motes might include serial or USB interfaces, allowing for direct connection to computers for programming and data extraction.

Berkeley Mote - Software Platform

⇒ Based on Tiny Micro-threading Operating System (TinyOS) which is designed for resource-constrained MEMS sensor. → Micro-Electro-Mechanical System Sensors

Challenges in Sensor Network Programming

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1. **Reliability:-** Due to their distributed nature and susceptibility to node and link failures, WSNs are inherently less reliable.
   → Programming environment must have
   1. Mechanisms for fault tolerance.
   2. Dynamic Topology Management.

2. **Resource Constraints:-** Sensor nodes are limited in processing power, memory and especially energy. Thus Programming tools must enable developers to create energy efficient applications that minimize power consumption for various operations.

3. **Scalability:-** WSNs can potentially include thousands of nodes, demanding programming models that scale effectively.
   → Tools should facilitate application developement for large networks.
   → Supporting self management and configuration to reduce the need for manual intervention.

4. **Data-Centric Networks:-** Programming models for WSNs need to support efficient data collection, aggregation, and analysis, possibly within the network itself to reduce redundancy and conserve energy.

Node-level Software Platforms

→ Provide Runtime environment, libraries and executing applications

→ Abstract the hardware details allowing developers to focus on appⁿ logic rather than low-level prog.

① TinyOS → An event driven OS
→ designed for low-power wireless devices
→ Allows reusable code (making it suitable for constrained environments of WSNs.

② Contiki → Lightweight, open-source OS for IoT.
→ powerful tools for building complex wireless systems.
→ Dynamic loading & unloading of individual prog. supported.
→ Flexible for deploying updates & new appⁿ

③ RIOT → Implements all relevant open std. supporting Internet of Things (IoT) applications.
→ Developer-friendly.
→ offers real time capabilities
→ Require minimal resources.

→ Remote internet of things

These platforms typically offer features such as multitasking, networking protocols optimized for WSNs, and libraries for accessing sensors and other hardware components.

Node - Level Simulators :-

→ Allows developer to model & test sensor N/w appⁿ in "Controlled environment" before deployment.

→ Crucial for evaluating the performance, reliability, and energy Consumption of appⁿ under various Conditions.

1 TOSSIM :-> Models the behaviour of sensor Networks at a high fidelity.
Tiny OS Simulator
→ Can simulate thousand of nodes by emulating the Tiny OS Code.
↳ lasting Support

2 Cooja : → Simulator for n/w of heterogenous nodes where each node can be individually configured.
Contiki OS Java Simulator.
→ For Contiki OS.

3 NS-3 : → Not exclusively for WSNs, NS-3 is a discrete n/w simulator supporting wide range of networking research, including 802.15.4 (used in WSNs), IPv6, and more.
n/w Simulator 3

Simulators play a vital role in the development lifecycle of WSN appⁿ by enabling the identifications and Correction of errors, performance bottlenecks and energy efficiency in early development process.
→ Helpful for testing n/w under extreme Condⁿ

+ Bottleneck: A situation that stops a process from progressing.

Security Issues in WSN

→ Security is a critical concern due to WSN's deployment in often unattended & hostile environments.

1) **Data Confidentiality** :- In WSNs, sensor nodes may collect and transmit critical data, making it imperative to protect this info from eavesdroppers.

2) **Data integrity** :- Integrity involves ensuring that the data collected and transmitted by sensor nodes is not altered or tampered with.

3) **Authentication** :- Authentication is crucial to verify the identity of nodes in the N/w to prevent unauthorized access and data falsification.

4) **Availability** :- Sensor n/w must remain operational even in the face of attacks or failures. Denial of Service (DoS) attacks, where attackers aim to exhaust the n/w resources or disrupt its operation, poses a significant threat to availability of WSN services.

5) **Secure Localization** :- Attackers might attempt to compromise the localization process by feeding false signals or manipulating the environment.

6) **Sybil Attacks** :- In a Sybil attack, a malicious node is illegitimately takes on multiple identities to damage the system of n/w.

7) **Node Capture Attacks** :- physically capturing of nods to extract sensitive info, e.g → cryptographic keys.

8) **Energy Drain Attacks** :- Depletions of nodes's energy reserves by targetting energy aspect by the attackers.

Future Research Topics/Direction in WSN...

Operating Systems

→ Investigate novel OS architecture, prog models, Scheduling algorithms, memory management, protection mechanism for WSNs.

→ Real-time appⁿ & other advanced features

Emerging Appⁿ Domains

→ Explore how WSNs can be customized for specialized appⁿ, such as precision agriculture, healthcare and smart cities.

Real Time Appⁿ

→ Investigate Real-time Commⁿ protocols, Scheduling algos, resource allocation techniques.

Energy efficient protocols

→ Develop novel energy efficient protocols that balance b/w accuracy & energy consumption.

Security & Privacy

→ Lightweight Security mechanisms suitable for resource-constrained sensor nodes.

Localization & Tracking

→ Develop robust localization algos that can handle noisy measurements & dynamic environments.

Integration with Edge & Fog Computing

→ Investigate seamless integration b/w WSN & edge/fog computing platforms to enhance data processing

As technology advances & new challenges arise, researchers will continue to shape the future of WSNs...

Thank You...

WSN Unit-4 & the Subject Semester Stuff ends with this...
Hope You liked the Notes :)

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Karan

Notes For WSN